



SAFETY DATA SHEET

POWER PAK LAUNDRY DETERGENT

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 파워 팩 세탁 세제
POWER PAK LAUNDRY DETERGENT

Other means of identification : Not applicable

Recommended use : Laundry detergent

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : No dilution information provided.

Company : ECOLAB KOREA LTD.
8F, Daemyung Tower
135 Beobwon-ro, Songpa-gu, Seoul Korea 05836
82-1588-7595

Emergency telephone : 82-1588-7595

Issuing date : 27.02.2017

2. HAZARDS IDENTIFICATION

GHS Classification

Skin sensitization : Category 1

GHS label elements

Hazard pictograms :



Signal Word : Warning

Hazard Statements : May cause an allergic skin reaction.

Precautionary Statements : **Prevention:**
Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray. Contaminated work clothing must not be allowed out of the workplace. Wear protective gloves.
Response:
IF ON SKIN: Wash with plenty of soap and water. If skin irritation or rash occurs: Get medical advice/ attention. Wash contaminated clothing before reuse.
Disposal:
Dispose of contents and container (according to the description in the related regulation).

Other hazards : None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
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Sodium Carbonate	497-19-8	80 - 90
Dodecylbenzenesulfonic acid, sodium salt	25155-30-0	1 - 10
c12-c14 pareth-7	68551-12-2	1 - 10
poly(oxy-1,2-ethanediyl), a-(2-propylheptyl)-w-hydroxy-	160875-66-1	1 - 10

All unlisted components do not contribute to the overall hazard of the product and are trade secret, making up 100% of the full composition.

4. FIRST AID MEASURES

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Wash off with plenty of water. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention if irritation develops and persists.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Remove to fresh air. Treat symptomatically. Get medical attention.
Protection of first-aiders	: If potential for exposure exists refer to Section 8 for specific personal protective equipment.
Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus
Special protective equipment for fire-fighters	: Use personal protective equipment.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions,	: Ensure clean-up is conducted by trained personnel only. Refer to
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protective equipment and
emergency procedures

protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for
containment and cleaning up : Sweep up and shovel into suitable containers for disposal.

7. HANDLING AND STORAGE

Advice on safe handling : Do not ingest. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray.
Use only with adequate ventilation. Wash hands thoroughly after
handling. Do not get in eyes, on skin, or on clothing.

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in
suitable labeled containers.

Storage temperature : 0 °C to 50 °C

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations
below occupational exposure standards.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : For prolonged or repeated contact use protective gloves.
Gloves should be discarded and replaced if there is any indication of
degradation or chemical breakthrough.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety
practice. Remove and wash contaminated clothing before re-use.
Wash face, hands and any exposed skin thoroughly after handling.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : powder

Color : opaque, blue

Odor : Floral

pH : 11.35, 1 %

Flash point : Not applicable

Odor Threshold : No data available

Melting point/freezing point : No data available

Initial boiling point and : > 100 °C

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boiling range

Evaporation rate : No data available

Flammability (solid, gas) : No data available

Upper explosion limit : No data available

Lower explosion limit : No data available

Vapor pressure : No data available

Relative vapor density : No data available

Relative density : 0.75

Water solubility : soluble

Solubility in other solvents : No data available

Partition coefficient: n-octanol/water : No data available

Autoignition temperature : No data available

Thermal decomposition : No data available

Viscosity, kinematic : No data available

Explosive properties : No data available

Oxidizing properties : The substance or mixture is not classified as oxidizing.

Molecular weight : No data available

VOC : No data available

10. STABILITY AND REACTIVITY

Chemical stability : Stable under normal conditions.

Possibility of hazardous reactions : No dangerous reaction known under conditions of normal use.

Conditions to avoid : None known.

Incompatible materials : None known.

Hazardous decomposition products : Decomposition products may include the following materials:
Carbon oxides
Nitrogen oxides (NO_x)
Sulfur oxides
Oxides of phosphorus

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes : Health injuries are not known or expected under normal use.

Skin : May cause allergic skin reaction.

Ingestion : Health injuries are not known or expected under normal use.

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Inhalation : Harmful if inhaled.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact : No symptoms known or expected.

Skin contact : Redness, Irritation, Allergic reactions

Ingestion : No symptoms known or expected.

Inhalation : No information available.

Toxicity

Product

Acute oral toxicity : Acute toxicity estimate : > 2,000 mg/kg

Acute inhalation toxicity : No data available

Acute dermal toxicity : No data available

Skin corrosion/irritation : No skin irritation

Serious eye damage/eye irritation : No eye irritation
Method: OECD Test Guideline 437
Test substance: Product

Respiratory or skin sensitization : No data available

Carcinogenicity : No data available

Reproductive effects : No data available

Germ cell mutagenicity : No data available

Teratogenicity : No data available

STOT-single exposure : No data available

STOT-repeated exposure : No data available

Aspiration toxicity : No data available

Ingredients

Acute dermal toxicity : c12-c14 pareth-7
LD50 Rabbit: > 2,000 mg/kg

12. ECOLOGICAL INFORMATION

Ecotoxicity

Environmental Effects : Harmful to aquatic life.

Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

Ingredients

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Toxicity to fish : Sodium Carbonate
96 h LC50 Lepomis macrochirus (Bluegill sunfish): 300 mg/l

Dodecylbenzenesulfonic acid, sodium salt
96 h LC50: 3.2 mg/l

c12-c14 pareth-7
96 h LC50 Fish: 1.5 mg/l

Ingredients

Toxicity to daphnia and other : Sodium Carbonate
aquatic invertebrates 48 h EC50 Ceriodaphnia (water flea): 213.5 mg/l

poly(oxy-1,2-ethanediyl), a-(2-propylheptyl)-w-hydroxy-
48 h EC50 Daphnia: 10 mg/l

Persistence and degradability

No data available

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

13. DISPOSAL CONSIDERATIONS

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations.

Do not contaminate ponds, waterways or ditches with chemical or used container.

Disposal considerations : Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (KR_DG)

Not dangerous goods

Sea transport (IMDG/IMO)

Not dangerous goods

Special precautions for user : None

15. REGULATORY INFORMATION

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National regulatory information

Regulation under the Occupational Safety and Health Act

Harmful Substances Prohibited from Manufacturing

Not relevant

Harmful Substances Required Permission for Manufacture

Not relevant

Hazardous substances requiring management

Not relevant

Controlled Substances Subject to Environment Monitoring

Not relevant

Controlled Substances Subject to Health Examination

Not relevant

Substances with Workplace Exposure Limits

Not relevant

Act on the Registration and Evaluation, etc. of Chemical Substances, Chemicals Control Act

Toxic Chemicals

Not relevant

Authorized chemicals

Not relevant

Restricted Chemicals

Not relevant

Prohibited Chemicals

Not relevant

Toxic Release Inventory

Not relevant

Accident Precaution Chemicals

Not relevant

Dangerous Substances Safety Management Act

Not Applicable to Dangerous Materials

Waste Control Act

Industrial waste

The ingredients of this product are reported in the following inventories:

United States TSCA Inventory :

On TSCA Inventory

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL

Australia Inventory of Chemical Substances (AICS) :

On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemical Substances :

On the inventory, or in compliance with the inventory

Japan. ENCS - Existing and New Chemical Substances Inventory :

On the inventory, or in compliance with the inventory

Japan. ISHL - Inventory of Chemical Substances (METI) :

not determined

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Korea. Korean Existing Chemicals Inventory (KECI) :
not determined

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :
On the inventory, or in compliance with the inventory

China. Inventory of Existing Chemical Substances in China (IECSC) :
On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet

Globally Harmonized System of Classification and Labeling of Chemicals (GHS)
IARC: (International Agency for Research on Cancer)
US. National Toxicology Program (NTP) Report on Carcinogens
ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

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Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.